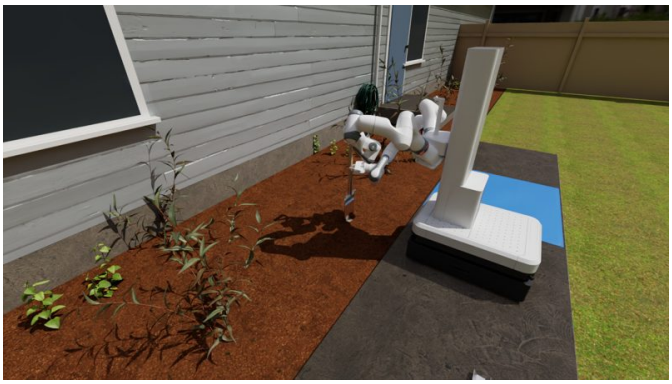
Dig soil (soil_excavation)

The soil excavation simulation: soil model, laboratory benches, demonstrations and interfaces

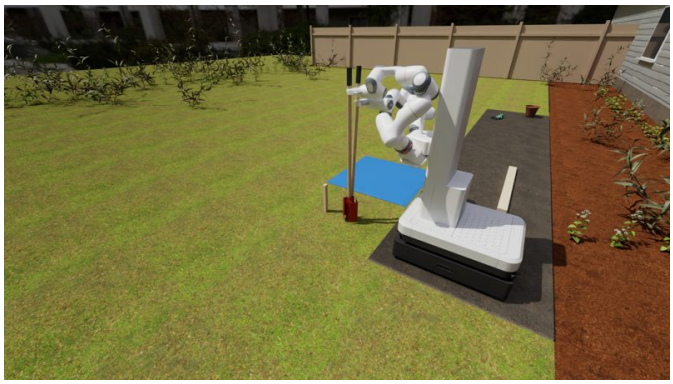
A front yard worked by the Franka Mobile FR3 Duo with real hand tools; the ground is a two-way coupled granular continuum (Newton 1.5 implicit MPM, 2 cm voxels, layered mulch, topsoil, turf, subsoil, yard soil, hardpan, sand) moved only by the tools, stones, root ball, post and the robot. Every assisted experiment is labelled; none is a policy action. The material table and the parameter ledger show the shipped solver settings. The laboratory benches, stability runs and demonstrations in sections 3 to 5 were run with an earlier solver configuration, including the direct-shear friction mapping mu = tan(phi)/0.546, a 1e-4 residual tolerance with a 50-iteration cap and an earlier soil-tool adhesion convention. They are not reruns with the shipped settings.

task	start state builds	start verdict (must fail)	failing criteria
penetrate-marked	yes	fails	edge_depth
spit-to-tarp	yes	fails	soil_on_tarp
planting-hole	yes	fails	depth_at_mark, top_diameter, spoil_on_tarp
plant-and-backfill	yes	fails	backfill_level, air_pockets, backfill_density
cable-trench	yes	fails	depth_coverage, width_coverage, spoil_on_tarp
post-hole	yes	fails	depth_at_stake, mid_depth_diameter, spoil_on_tarp
set-post	yes	fails	backfill_level, post_plumb
break-hardpan	yes	fails	loosened_80mm, fragments_on_tarp
auger-holes	yes	fails	hole_1, hole_2, hole_3, hole_4, hole_5
backfill-pipe	yes	fails	haunch_voids, sand_cover, surface_at_grade
expose-line	yes	fails	crown_exposed

1. The yard and every task's start state



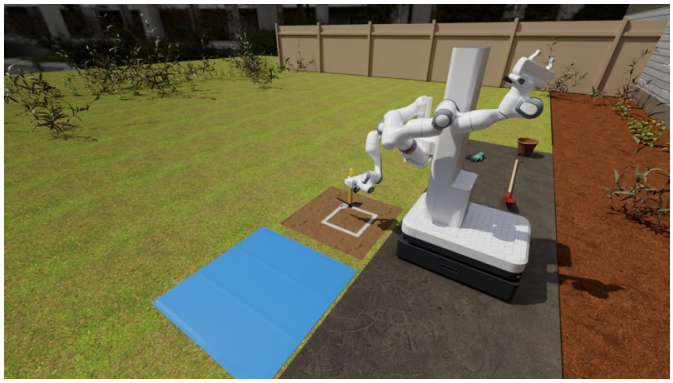
bed preset: the robot on the walkway, spade at the mark



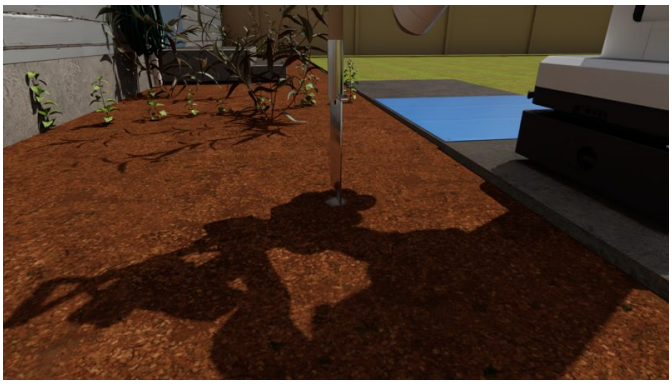
post preset: lawn, clamshell digger at the stake



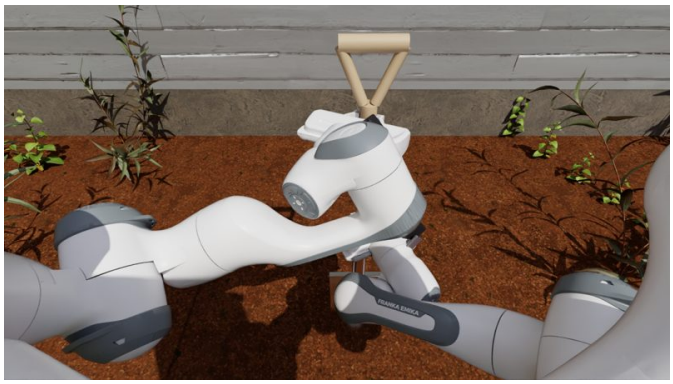
irrigation preset: 811 flags and paint over the lateral



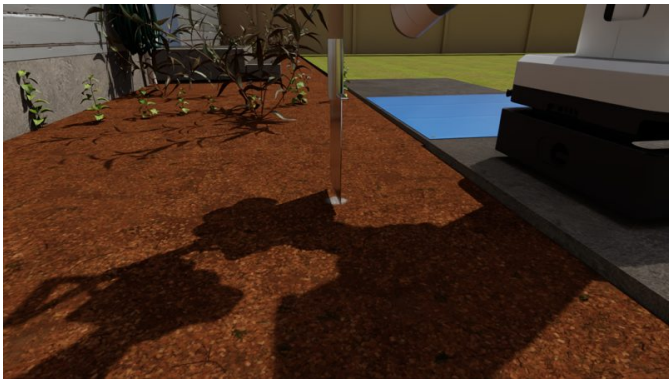
hardpan preset: path corner, hand mattock



penetrate-marked: work view (diagnostic camera)



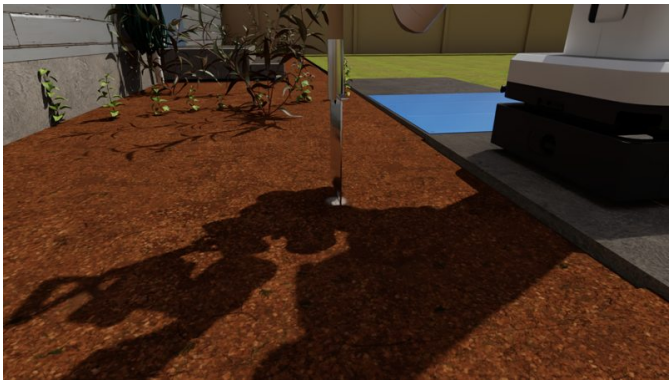
penetrate-marked: head_rgb (the robot's camera)



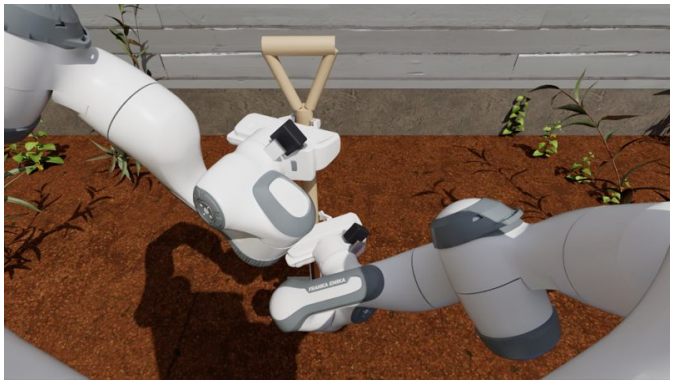
spit-to-tarp: work view (diagnostic camera)



spit-to-tarp: head_rgb (the robot's camera)



planting-hole: work view (diagnostic camera)



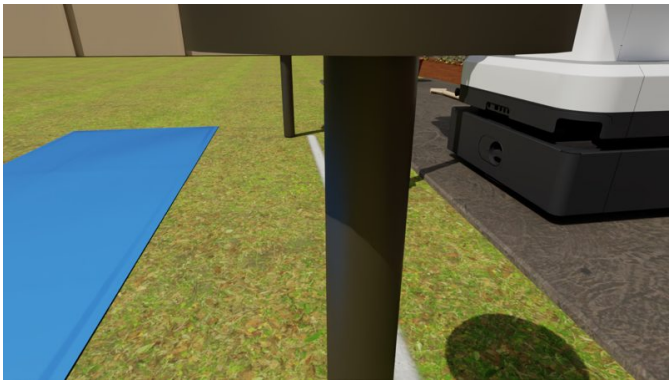
planting-hole: head_rgb (the robot's camera)



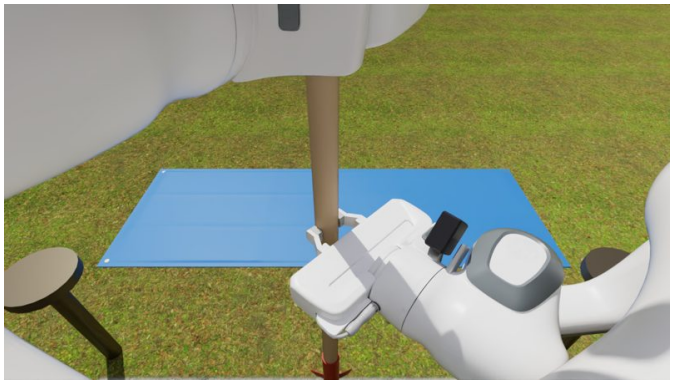
plant-and-backfill: work view (diagnostic camera)



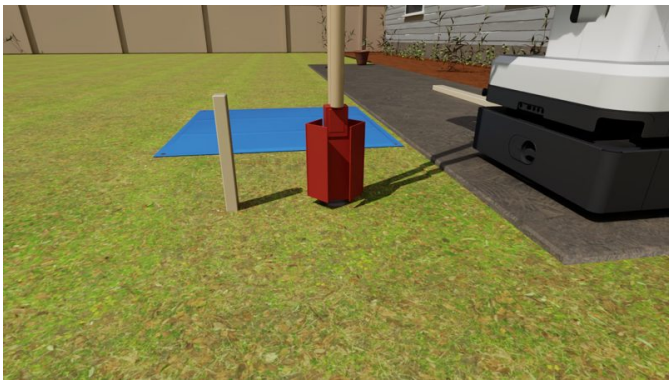
plant-and-backfill: head_rgb (the robot's camera)



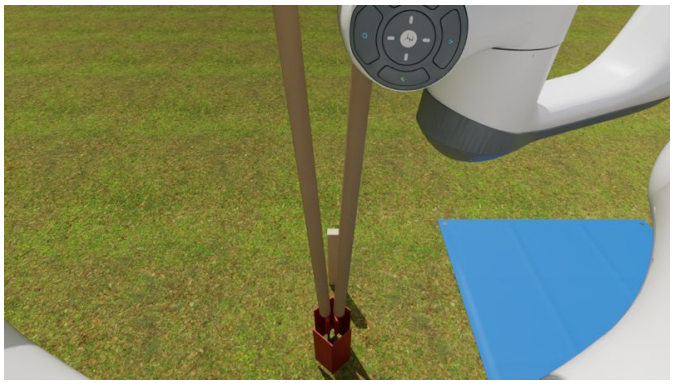
cable-trench: work view (diagnostic camera)



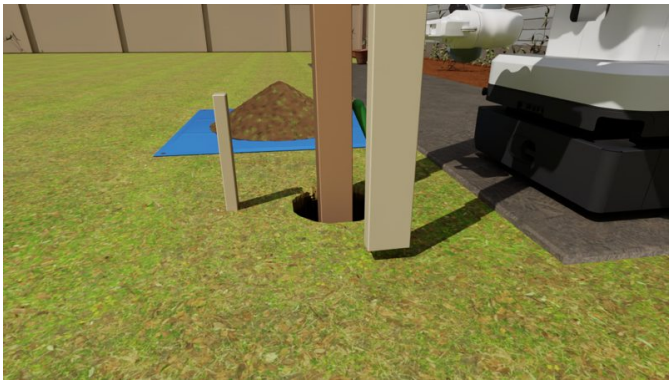
cable-trench: head_rgb (the robot's camera)



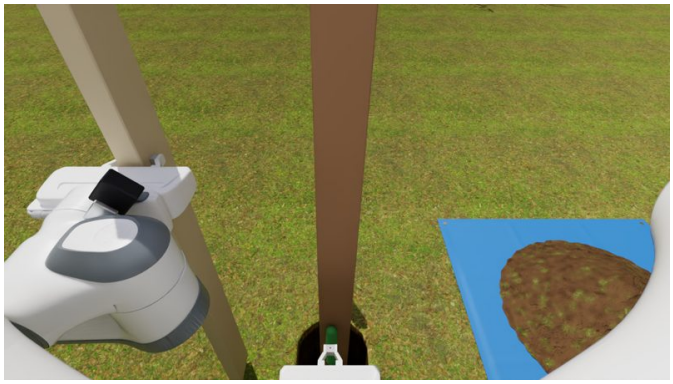
post-hole: work view (diagnostic camera)



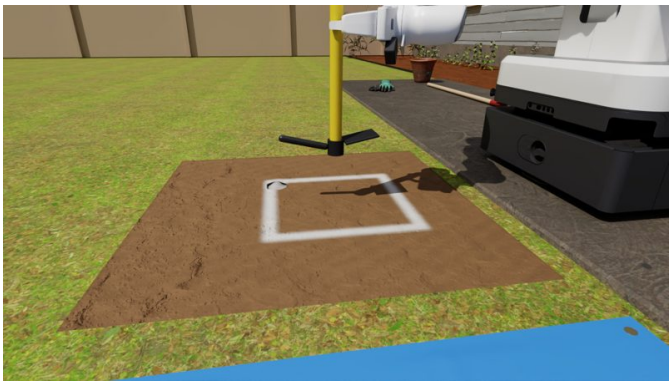
post-hole: head_rgb (the robot's camera)



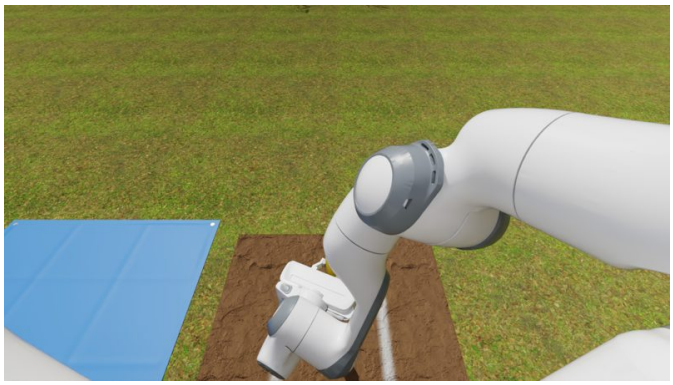
set-post: work view (diagnostic camera)



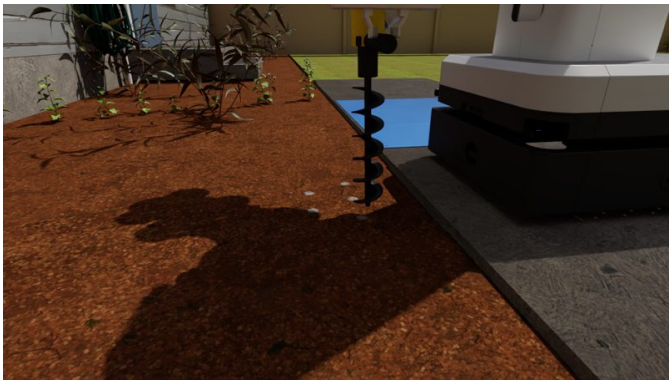
set-post: head_rgb (the robot's camera)



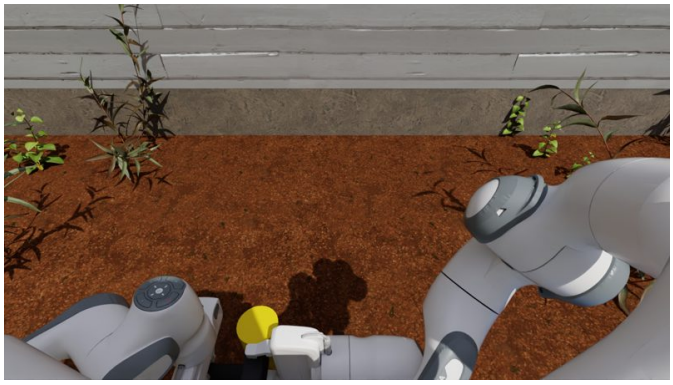
break-hardpan: work view (diagnostic camera)



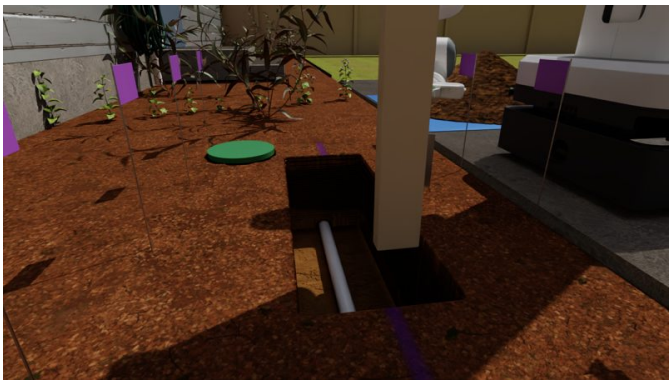
break-hardpan: head_rgb (the robot's camera)



auger-holes: work view (diagnostic camera)



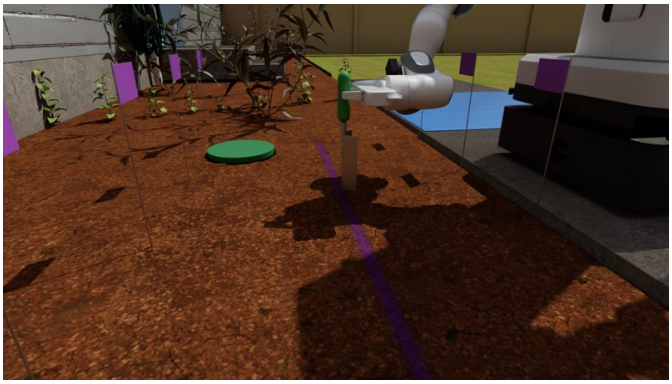
auger-holes: head_rgb (the robot's camera)



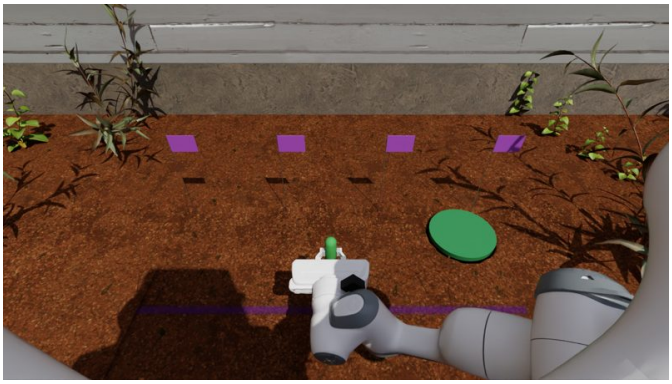
backfill-pipe: work view (diagnostic camera)



backfill-pipe: head_rgb (the robot's camera)



expose-line: work view (diagnostic camera)



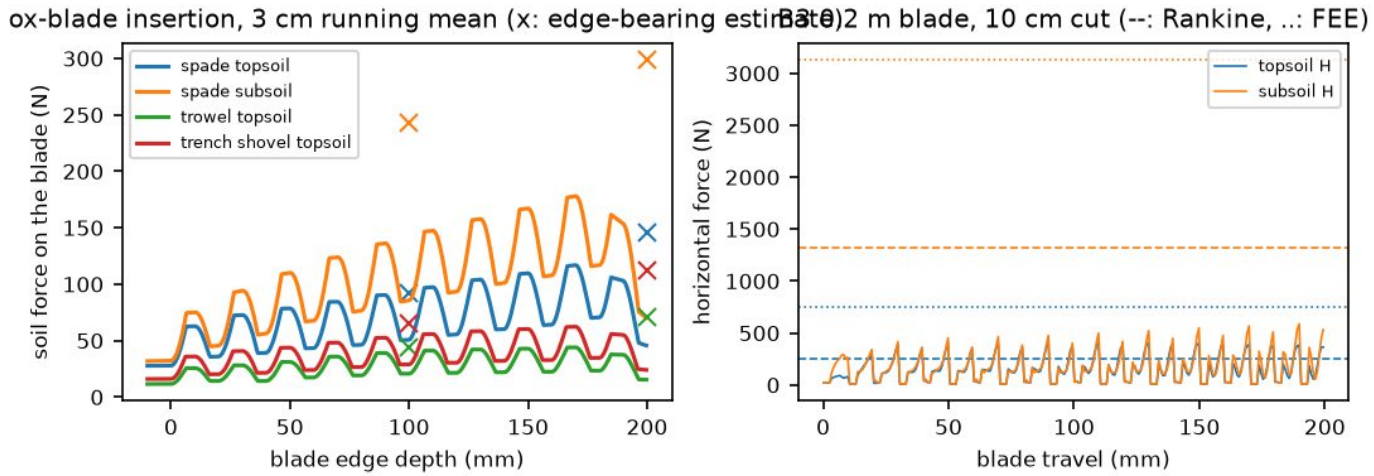
expose-line: head_rgb (the robot's camera)

2. Soil model

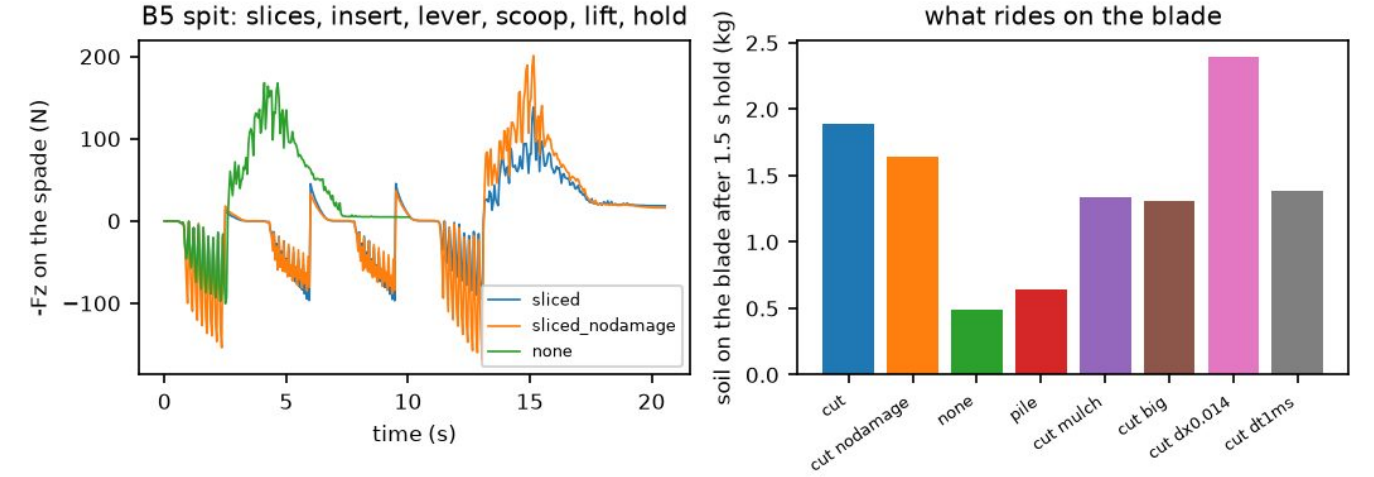
Yield (Newton 1.5 implicit MPM, capped Drucker-Prager): sqrt(J2) <= s h(J_p) + mu (p + p_t), cap Y h(J_p), tension p_t = r_t Y h(J_p), hardening h = sinh(-H ln J_p), J_p updated by the plastic volume. Bank soil starts at J_p0 with h = 1; loosening by its swell ratio drops h to 0.15 (cohesion and tension lost, friction kept); compaction restores it. Friction and cohesion follow the compression-meridian Mohr-Coulomb match: mu = 6 sin(phi)/(sqrt(3)(3 - sin(phi))), s = 6 c cos(phi)/(sqrt(3)(3 - sin(phi))) - mu p_t. The solver's critical fraction 0.65 keeps loose spoil (packing 0.78) jammed. Blade cut closure: a moving blade plate, pick or adze remolds the soil within half a voxel of its surface to the loose state (a 2.5 mm cut cannot stay open on a 2 cm grid; the auger is excluded). Coupling every 2 ms: driven and heavy bodies are kinematic in the soil with their reaction applied over the next 4 MuJoCo ticks plus an in-interval damper; light free bodies are dynamic colliders.

material	rho	c target (Pa)	phi (deg)	solver mu	solver s (Pa)	J_p bank	H
topsoil	1300.0	3000.0	32.0	0.7432	3196.4	0.7428	2.965
turf	1150.0	9000.0	32.0	0.7432	8474.4	0.7428	2.965
subsoil	1700.0	20000.0	26.0	0.5928	21344.8	0.7428	2.965
yard	1650.0	12000.0	28.0	0.6427	12897.6	0.7428	2.965
hardpan	1800.0	45000.0	28.0	0.6427	47964.3	0.6967	2.439
sand	1600.0	500.0	34.0	0.7936	588.3	1.0	0.0
gravel	1550.0	0.0	40.0	0.9446	0.0	1.0	0.0
mulch	350.0	200.0	40.0	0.9446	36.2	1.0	0.0

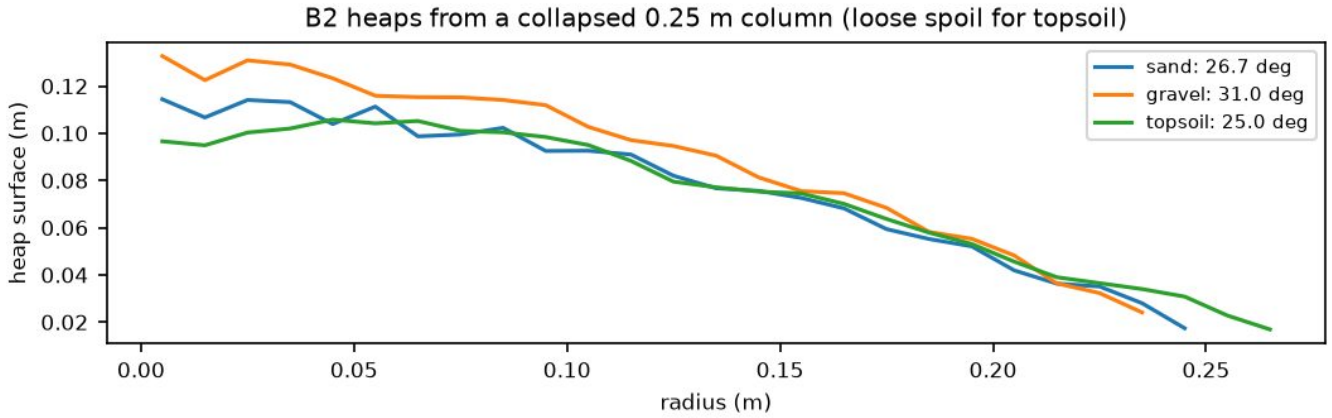
3. Virtual soil laboratory (rigid fixtures only)



B4 tool insertion and B3 blade cutting on the production soil path

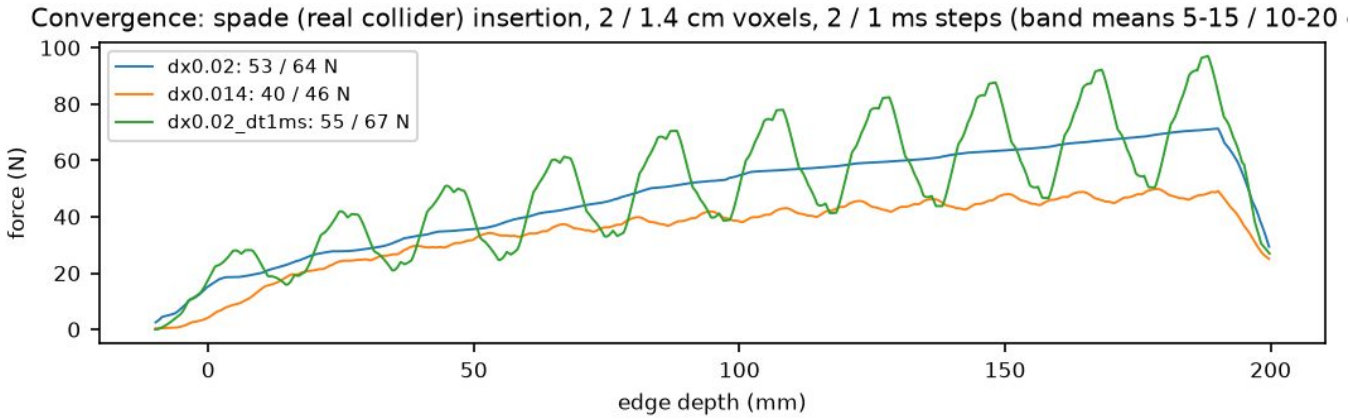


B5 spit lift: three slices, insertion, lever, scoop and lift of a kinematic spade in topsoil; the cut closure on/off and a loose spoil heap



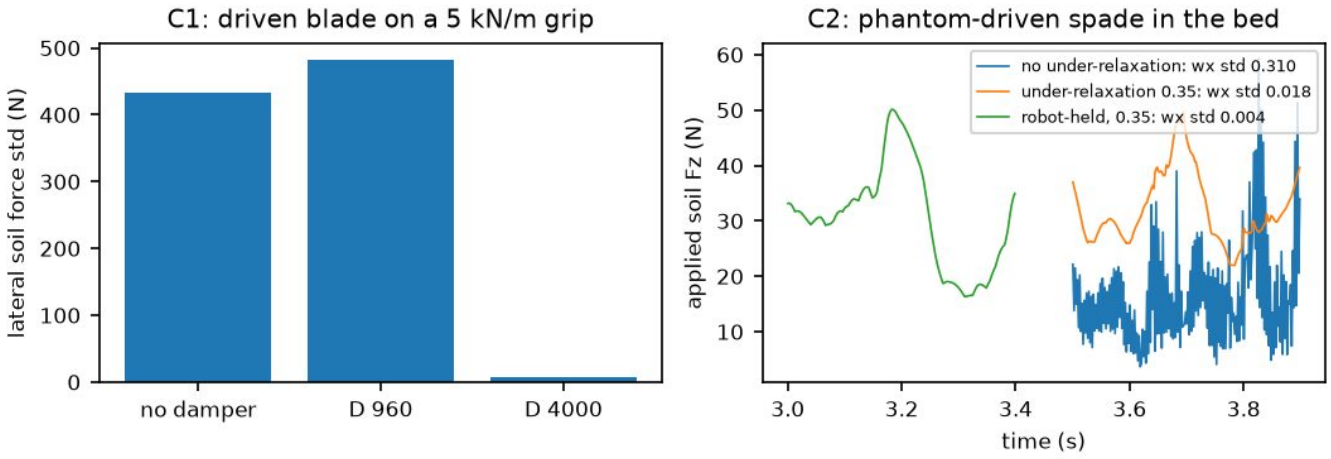
B2 heap profiles

bench	measured	reference
loose topsoil heap	25.0 deg	spoil 30-40 deg
sand heap (c = 0)	26.7 deg	33-36 deg static
gravel heap	31.0 deg	38-42 deg
swell of dumped topsoil	22 %	CAT 25-56 %
0.3 m cut, sand	stands: True, slumped 0.0 kg	Hc 0.24 m: fails
0.3 m cut, topsoil	top edge -3.7 mm, toe 0.0 kg	Hc 1.7 m: stands

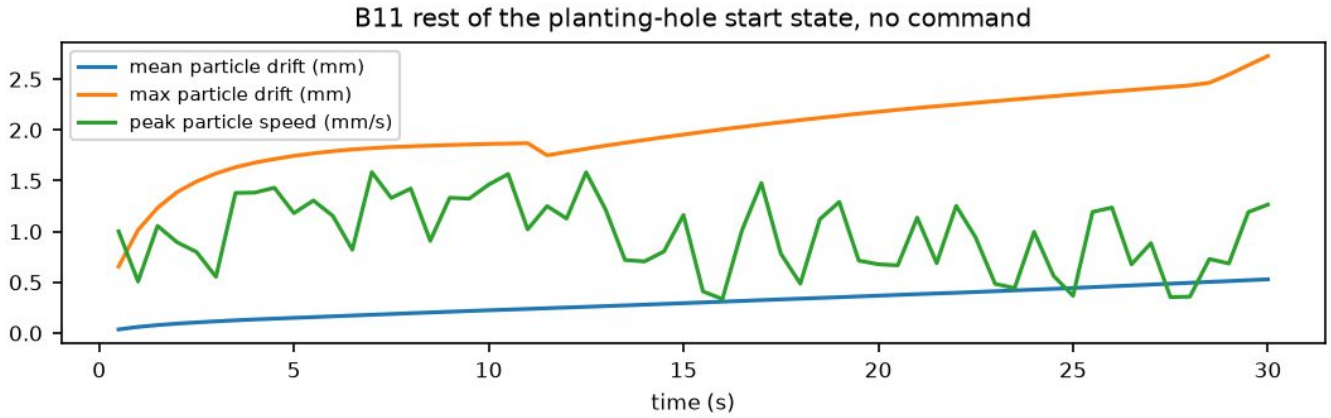


Convergence of the insertion force (voxel and soil step)

4. Coupling and stability



C1: a 0.6 kg blade on a 5 kN/m grip in the soil, explicit lag without and with the damper

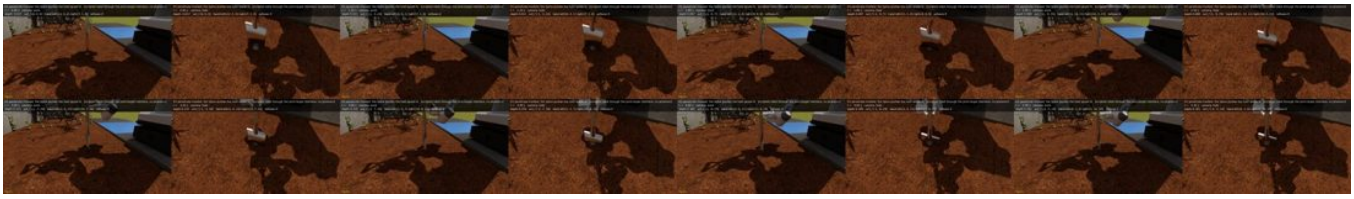


B11 rest

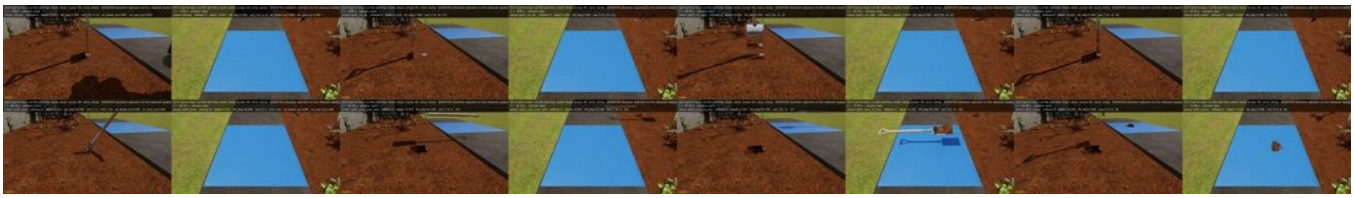
bench	result
drop	spade at -0.086 m, tilt 0.93 deg, fault None
ram 1 m/s	final z -0.166 m, fault None
off the slab	base [-0.0267, -0.2285, 1.5483], fault None
post, loose backfill	430 mm at 150 N
post, tamped backfill	202 mm at 150 N
post time history (B10)	held 150 N: 125-141 mm, backfill h(J_p) median 0.98 of bank (soft, not failing)
150 mm hole, bank topsoil (B12)	stands True after 4 s
same, 1 cm remolded wall (B12)	stands True after 4 s
soil at rest in a bin, 90 s (B11b)	mean settlement -7.0 mm, late rate -0.049 mm/s

5. Demonstrations (robot-scripted and phantom-assisted)

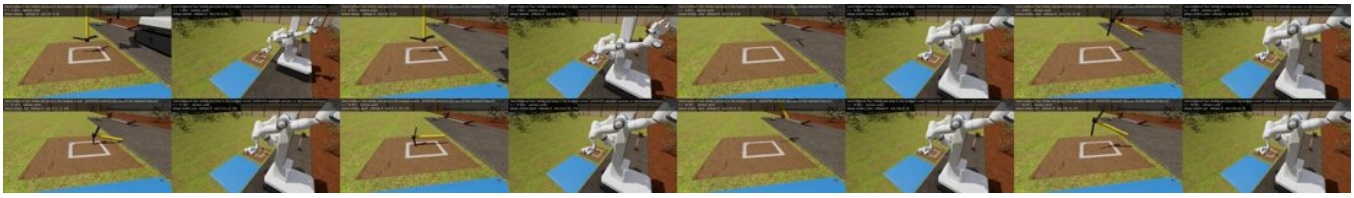
demo	assistance	outcome	sim (s)	wall/sim
D1 robot pushes the spade in at the mark	scripted robot through the public joint-target interface (ToolDriver.s	evaluator success; no_fault=null, protective_stops=0.0, edge_depth=0.181, edge_on_mark=0.0047	5.5	95x
D2-P spits cut, levered, lifted and dumped on the tarp	ASSISTED phantom operator on the released spade (PosePhantom 20 kN/m,	lifted_kg [1.2213499546051025]; on_tarp_kg 0.9525500535964966	46.6	34x
mattock strikes break the hardpan	ASSISTED phantom operator on the released mattock (strikes ~2-3 m/s he	loosened_kg 3.988799571990967	19.5	59x
post-hole digger jab, close, lift	ASSISTED phantom operator on digger half a + a labelled 200.0 N m hing	on_tarp_kg 1.9424998760223389; plugs_kg [0.8257499933242798, 1.4156999588012695,	44.7	27x
trowel backfill from the spoil heap	ASSISTED phantom operator on the released trowel	on_tarp_kg 13.129098892211914; added_to_hole_kg [0.33065032958984375, 0.4259490966796875	41.3	71x
2x4 tamping around the post	ASSISTED: hole backfilled loose by construction (B10), phantom operato	density_before_kg_m3 515.3104858398438; density_after_kg_m3 26.56681251525879	14.7	37x
strike log: gentle touch vs stab on the pipe	ASSISTED phantom operator on the released trowel (gentle touch at 3 cm	strikes_after_gentle 0	7.7	43x
D9 robot drills with the auger	scripted robot through the public interface (gripper force on the pist	evaluator not met; no_fault=null, protective_stops=0.0, hole_1={"depth": 0.085, "diameter": 0, hole_2={"depth": 0.063, "diameter": 0	16.2	46x
D2 the robot's own spit attempt	scripted robot through the public joint-target and base-twist interfac	evaluator not met; no_fault=null, protective_stops=1.0, soil_on_tarp=0.0, spill_fraction=0.0	6.7	92x



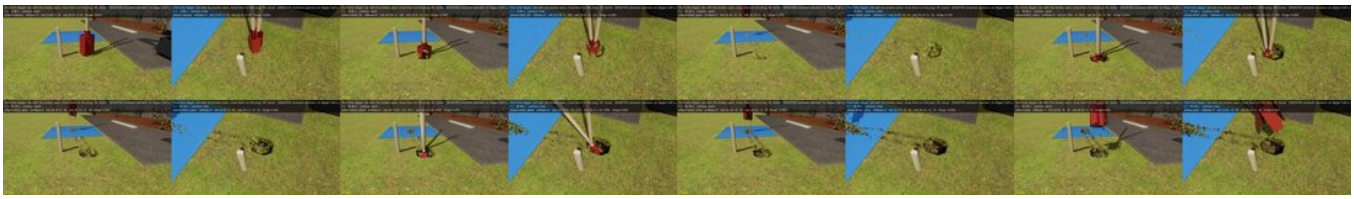
D1 robot pushes the spade in at the mark: frames across the whole video



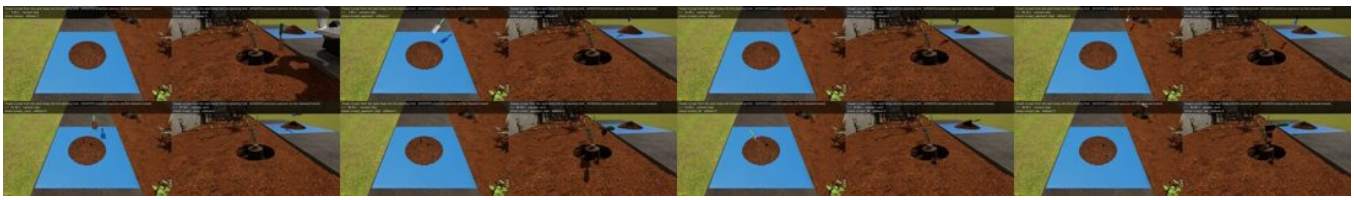
D2-P spits cut, levered, lifted and dumped on the tarp: frames across the whole video



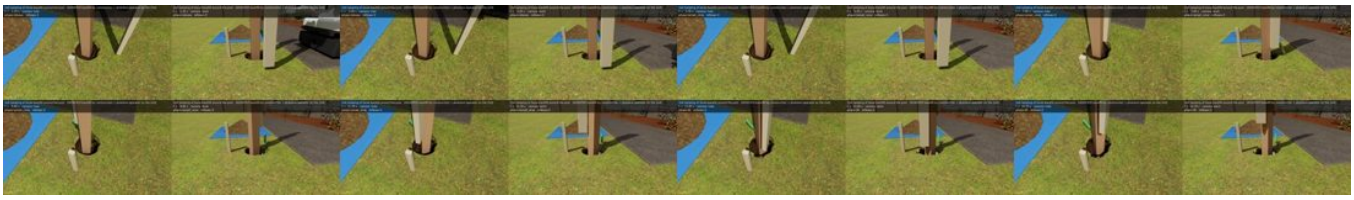
mattock strikes break the hardpan: frames across the whole video



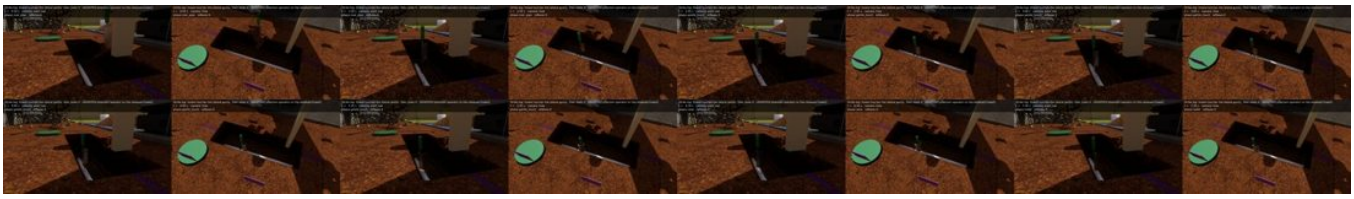
post-hole digger jab, close, lift: frames across the whole video



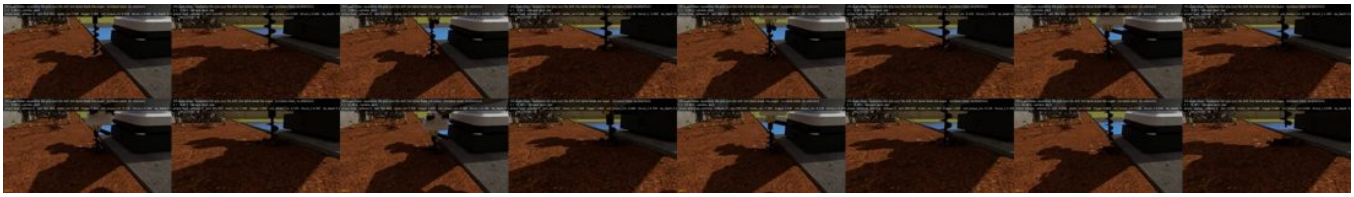
trowel backfill from the spoil heap: frames across the whole video



2x4 tamping around the post: frames across the whole video



strike log: gentle touch vs stab on the pipe: frames across the whole video



D9 robot drills with the auger: frames across the whole video

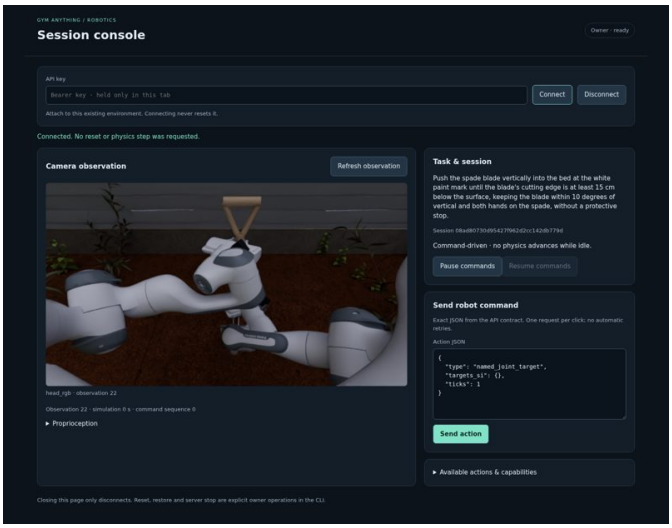
6. Evaluator, runtime and interfaces

task	case	succeeded	failing
spit-to-tarp	start	False	soil_on_tarp
spit-to-tarp	pass	True	
spit-to-tarp	near_miss: 1.2 kg on the tarp	False	soil_on_tarp
planting-hole	start	False	depth_at_mark, top_diameter, spoil_on_tarp
planting-hole	pass	True	
planting-hole	near_miss: 0.13 m deep	False	depth_at_mark
planting-hole	near_miss: a third of the spoil on the walkway	False	spoil_on_tarp, walkway_clean
cable-trench	start	False	depth_coverage, width_coverage, spoil_on_tarp
cable-trench	pass	True	
cable-trench	near_miss: 0.14 m deep	False	depth_coverage
post-hole	start	False	depth_at_stake, mid_depth_diameter, spoil_on_tarp
post-hole	pass	True	
post-hole	near_miss: 0.52 m deep	False	depth_at_stake
break-hardpan	start	False	loosened_80mm, fragments_on_tarp
break-hardpan	pass	True	
break-hardpan	near_miss: loosened 0.06 m	False	loosened_80mm
auger-holes	start	False	hole_1, hole_2, hole_3, hole_4, hole_5
auger-holes	pass	True	
auger-holes	near_miss: hole 1 drilled 35 mm off its mark	False	hole_1
expose-line	start	False	crown_exposed
expose-line	pass	True	
expose-line	near_miss: 20 mm of soil left on the crown	False	crown_exposed

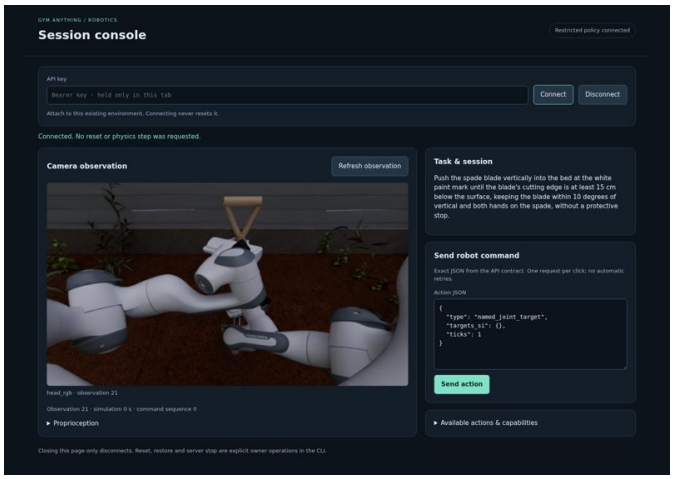
check	passed
dashboard_page_and_script_200	True
dashboard_calls_200	True
camera_png	True
policy_reads_200	True
policy_actions_200	True
policy_checkpoint_forbidden	True
policy_state_forbidden	True
owner_save_restore_200	True
restore_returns_tick	True
worst_ratio_below_200	False
one_tick_command_refused	True
batch_below_200	True
trace_route_200	True
export_with_traces_ok	True
variations_supported	True
variation_start_200	True
evaluation_under_client_timeout	True
gar_verify_cli_ok	True

command	sim (s)	wall (s)	wall/sim	status
base_twist	0.5	77.269	154.5	200
base_twist	0.5	101.62	203.2	200
named_joint_target	0.5	103.717	207.4	200
gripper	0.5	102.116	204.2	200
cartesian_delta	0.5	101.101	202.2	200
base_twist (shortest legal)	0.1	55.285	552.8	200
named_joint_target (shortest legal)	0.1	56.325	563.2	200

command	sim (s)	wall (s)	wall/sim	status
gripper (shortest legal)	0.1	55.492	554.9	200
cartesian_delta (shortest legal)	0.1	55.946	559.5	200
cartesian_delta (shortest legal)	0.1	55.526	555.3	200
keyboard (shortest legal)	0.1	57.117	571.2	200
base_twist (shortest legal)	0.1	81.829	818.3	200
keyboard UP after the stop's base hold	0.1	58.068	580.7	200



dashboard, owner key



dashboard, policy key

7. Parameter ledger

name	value	unit	label	basis
SOIL_TICKS	4	robot ticks	derived	soil step 2 ms = 4 x 0.5 ms: a tool at the FR3's 2 m/s Cartesian limit moves 4 mm = 0.2 voxel per soil step; the coupling lag equals one soil step
VOXEL	{"yard": 0.02, "fine": 0.01}	m	derived	2 cm yard blocks (19.6 ms per loaded step for 512 k particles), 1 cm for the auger bed and the physics benches
PPC_AXIS	2	-	repo	2 x 2 x 2 = 8 particles per cell (Newton MPM examples)
MPM_ITERATIONS	1000	-	numerical qualification	Nonlinear iteration cap, paired with 1e-6 residual tolerance. Increasing the cap alone at 1e-4 does not stop drift; tightening the tolerance gives late loose-column rates of 0.050 mm/s at 200 iterations and 0.024 mm/s at 1000, against 0.821 mm/s at 1e-4. Long-horizon and loaded-world qualification remain separate; the cap is not a soil property.
MPM_TOLERANCE	1e-06	-	numerical qualification	Newton stress residual tolerance. A 1e-4 tolerance leaves continuing particle and inclusion drift even at a 1000-iteration cap. Tighter convergence costs more and must be included in runtime measurements.
CRITICAL_FRACTION	0.78	-	derived	packing fraction below which the yield surface collapses (Newton critical_fraction): disturbed common earth bulks 20-50 %, typ. 25 %, loam 23 % [CAT swell tables]; this yard uses amended loam, not the 43-56 % organic topsoil category; 1/0.78 = 28 % swell of bank material packed at fraction 1.0 (B6 checks the loose density of real spoil)
SOLVER_CRITICAL_FRACTION	0.65	-	calibrated	Newton's critical_fraction (below it a node's yield surface collapses: free flow). Loose spoil sits at CRITICAL_FRACTION (0.78, its 28 % swell); with the solver's threshold also at 0.78 every dilation of loose soil unjams it and spoil heaps slump to 11-15 deg (B2); 0.65 leaves loose soil jammed until it dilates past 54 % swell
COLLIDER_BASIS	"S2"	-	calibrated	Newton 1.5's default quadratic serendipity collider basis: its contact activation band is half a voxel / degree = 5 mm at 2 cm voxels. With trilinear Q1 (1 cm band) a 2.5 mm spade edge acted 2 cm thick: insertion 193 / 334 N at 10 / 20 cm in topsoil and 822 / 1328 N in subsoil against the edge-bearing estimates 93 / 146 and 243 / 299 N; S2 gave 98 / 188 N (B4). S2 costs 2.5-3x
GRID_CAPACITY	2.0	x material cells	authored	rebuildable sparse grid capacity over the cells the particles fill at bank packing: the one-cell halo, loose spoil on the tarp, heave and airborne clods (a planting hole moves ~3 % of the bed block; overflow faults the episode). 3.0 holds 6.9 GB of the shared GPU for the bed block
MAX_SOIL_SPEED	15.0	m/s	authored	fault guard: no clod in a garden moves faster
SOIL_SETTLE_S	0.3	s	authored	gravity settle of the lattice before tick 0 (elastic compression gamma z / E < 0.3 %); B11 measures the rest drift after it
MATERIALS	{"topsoil": {"water": "loam", "rho": 1300.0, "c": 3000.0, "phi": 32.0, "pt": 500.0, "E": 5000000.0, "nu": 0.3,	kg/m3, Pa, deg, Pa, Pa, -, Pa, -, -	published / estimate	per-material basis
MOISTURE	{"dry": {"ca": 0.3, "mut": 0.9}, "moist": {"ca": 1.0, "mut": 1.0}, "wet": {"ca": 2.0, "mut": 0.8}}	x	estimate	episode moisture state scales the soil-tool interface: adhesion and interface friction follow Birch et al.'s soil-metal water-content trends [BI] (adhesion peaks wet, friction falls). The soil's own cohesion follows MOISTURE_COHESION per soil class
MOISTURE_COHESION	{"clay": {"dry": 1.8, "moist": 1.0, "wet": 0.5}, "loam": {"dry": 0.6, "moist": 1.0, "wet": 0.6}, "sand": {"dry	x	estimate	cohesion and tension cut-off relative to the moist value, per soil class (MATERIALS 'water'). Clay soils (subsoil, yard soil, hardpan) gain strength as they dry: shrinkage and suction harden them (a dry compacted clay path is the hardpan's own basis, cone index >= 2-3 MPa [PSU]; unsaturated soil mechanics, Fredlund & Rahardjo 1993) and soften when wet. A friable loam's apparent
YIELD_LAW	{"A": 1.87, "B": 1.04, "mode": "compression_meridian"}	-	derived	Exact ascending-branch mapping from the installed Newton stress projector: sqrt(J2) <= s + mu(p+pt). Compression-meridian MC match, checked by B15. A/B serve only optional modes; the default does not use them. Constitutive identity is not excavation calibration.
KAPPA_DS	0.546	-	calibrated / unresolved	Scale mu = tan(phi) / KAPPA_DS of the optional direct-shear mapping, from a B1 outer-cone run and used for heap tuning; B1 on the default mapping misses its friction-angle target. Not a verified material constant.
KAPPA_DS_C	0.922	-	calibrated / unresolved	B1 intercept scale of the optional direct-shear mapping: c_ds = KAPPA_DS_C (s + B mu p_t). B1 on the default mapping gives 6.9 kPa against 3 kPa. Calibration across stress paths remains open.
E_CAP	40000000.0	Pa	derived	Young's modulus used by the solver = min(E, E_CAP): no material is capped (the largest estimated modulus in MATERIALS, hardpan, is 40 MPa). Without the in-interval damper and the under-relaxation a held blade chatters on the explicit 2 ms coupling at 5 MPa; with both, the robot-held spade pushed into the bed is as steady at 20 MPa as at 1 MPa (velocity noise <= 0.008 m/s, soil
CUT_CLOSURE	{"band_voxels": 0.5, "v_min": 0.01}	voxels, m/s	closure	a moving blade plate, pick or adze remolds the soil it passes through to the loose state (J_p >= J_p loose: residual strength retained, friction kept) within half a voxel across its thickness only, advancing edge-first; the 2 cm grid cannot keep a 2-3 mm cut open, so without it cut faces re-bond (B5, D2-P). Not applied to the auger: with it the turning flight loosens its whole
COUPLING_RELAXATION	0.35	-	calibrated	under-relaxation of the soil wrench applied to kinematically coupled bodies (running average per 2 ms soil step): removes the period-2 oscillation of the explicit coupling (D2-P diagnostics)
ROBOT_GUARD	{"below_grade_m": 0.25, "speed_m_s": 4.0}	m, m/s	authored	fault guard on the robot's own bodies (the classroom guard covers objects only): no body of the TMR base (hull, drive units, casters; arms excluded, a hand may reach into a dug hole) goes 0.25 m below grade on a yard whose ground is everywhere at grade or above, and no FR3 link (1.7-2 m/s TCP limit) or base (~1 m/s) moves at 4 m/s
COUPLING_DAMPING	{"v_ref": 0.02, "w_ref": 0.1, "cap": 1.0}	m/s, rad/s, -	calibrated	a kinematically coupled body's lagged soil wrench is supplemented, inside each 2 ms soil interval, by a damper on its deviation from the velocity the soil step assumed: D = min(|F_soil| / v_ref, cap m / dt_mujoco) (rotation likewise with w_ref and the inertia). It vanishes in steady motion and in air. A 0.6 kg blade on a 5 kN/m grip chatters at +-434 N std without it and holds
LOOSE_STRENGTH	0.15	x bank	estimate	cohesion, tension cut-off and cap of loosened cohesive soil relative to the bank (remoulded loam keeps a little suction cohesion; the cap keeps the friction cone of loose spoil up to ~0.15 Y, i.e. 18 kPa for topsoil, piles to ~1 m: B2)
HARDENING_RATE	1.0	-	authored	J_p follows the plastic volume ratio in compaction
SOFTENING_RATE	1.0	-	authored	J_p follows the plastic volume ratio in dilation: a loosened cell loses its cohesion as it bulks toward the loose density

name	value	unit	label	basis
INTERFACE	{"stainless": [0.4, 750.0], "steel": [0.47, 1500.0], "aluminium": [0.42, 1000.0], "wood": [0.5, 500.0], "pvc":	mu, Pa	estimate	smooth steel in moist loam delta 20-30 deg; adhesion means 1.74 / 2.35 / 3.35 kPa (sandy loam / clay loam / clay) at high water contents and 0.22-7.62 kPa range [BI]; polished stainless 'for minimal soil adhesion' [SJ]; Vulkollan drive tread uses the robot model's 0.6 traction estimate; a free rolling caster uses 0.1 resistance; tires have no cohesive adhesion
FAR_FIELD	"rough"	-	authored	block side/bottom faces: mu = tan(phi), adhesion = c of the adjacent soil, standing in for undisturbed ground
SPADE	{"head": [0.18, 0.29, 0.0025], "length": 0.985, "mass": 1.9, "shaft_r": 0.0175, "tread": 0.012, "edge_sagitta"	m, kg	datasheet	Spear & Jackson Traditional Stainless digging spade: head 290 x 180 mm, mirror-polished stainless, treaded, hardwood D-handle; Neverbend 1160SP 985 mm overall, 1.9 kg [SJ]
TRENCH_SHOVEL	{"head": [0.108, 0.292, 0.002], "length": 1.49, "mass": 1.6, "shaft_r": 0.016, "v_deg": 20.0, "lift_deg": 18.0	m, kg	datasheet	Ames Razor-Back 47171: 11.5 x 4.25 in V blade, 48 in handle, 58.75 in overall, 3.5 lb [AM1]; V and lift angles estimated from product photos
TROWEL	{"head": [0.078, 0.15, 0.0025], "length": 0.307, "depth": 0.03, "mass": 0.165, "grip_r": 0.016}	m, kg	datasheet	Fiskars Ergo trowel: 307 mm long, 78 mm wide, 30 mm deep, cast aluminium, 0.165 kg [FI]
MATTOCK	{"handle": 0.381, "head_mass": 0.68, "mass": 0.9, "pick": 0.15, "adze": [0.075, 0.09]}	m, kg	datasheet + estimate	garden mattock: 1.5 lb forged pick + adze head, 15 in fiberglass handle [IT]; 0.9 kg all-in (estimate: a comparable 15 in pick lists 1.49 lb)
DIGGER	{"blade": [0.14, 0.203], "spread": 0.14, "handles": 1.22, "pin_above_tip": 0.22, "mass": 3.2, "pin_friction":	m, kg, N m	datasheet + estimate	Ames/True Temper clamshell post-hole digger: 8 x 5.5 in blades, 5.5 in spread, 48 in handles, 58.5 in overall, 6.75-7.2 lb [AM2]; pin height above the tips and pin friction estimated (no published value)
DRILL	{"mass": 2.1, "side_handle": 0.15, "gears_rpm": [450.0, 1300.0, 2000.0], "torque_hard": 95.0, "torque_soft": 6	kg, rpm, N m, W, m, (N, N/m)	datasheet + estimate	DeWalt DCD996 18/20 V XR: 0-450 / 0-1300 / 0-2000 rpm, 95 N m hard / 66 N m soft, 820 W, 2.1 kg with battery, 360 deg side handle [DW]; trigger travel and spring estimated: 30 N preload, 42 N at full travel. Known limitation: in the start hold the drill hangs from the side handle and its weight rests ~37 N on the open right hand's finger at the trigger, which keeps it pressed;
DRILL_CLUTCH	{"1": 1.0, "2": 1.9, "3": 2.8, "4": 3.7, "5": 4.6, "6": 5.5, "7": 6.4, "8": 7.3, "9": 8.2, "10": 9.1, "11": 10	N m at the spindle	estimate	DCD996 electronic clutch, 11 settings + drill mode (no cap); slip torques are not published, linear 1-10 N m assumed (typical driver clutches span ~1-10 N m); None = drill mode
AUGER	{"diameter": 0.0762, "length": 0.305, "flight_len": 0.28, "pitch": 0.057, "shaft_r": 0.0079, "flight_t": 0.003	m, kg	datasheet + estimate	Power Planter 312: 3 x 12 in, 11 in digging depth, 5/8 in solid shaft, 10-gauge flighting, 3/8 in hex, 0.71 kg [PP]; pitch 0.75 D estimated from product photos
TAMPER	{"section": [0.038, 0.089], "length": 1.2, "rho": 450.0}	m, kg/m3	published + estimate	2x4 SPF actual 1.5 x 3.5 in; SPF ~450 kg/m3 (estimate) -> 1.83 kg
ROOTBALL	{"diameter": 0.165, "height": 0.16, "mass": 2.0}	m, kg	published + estimate	#1 nursery container (ANSI Z60.1 2.49-4.12 L; 6-7 in pots [Z60]); wet peat-bark mix 0.5-0.65 kg/L
POST	{"section": 0.0889, "length": 1.829, "rho": 370.0}	m, kg/m3	published + estimate	4x4 (3.5 in actual) western red cedar, 6 ft; ~370 kg/m3 air-dry (estimate) -> 5.35 kg
PIPE	{"od": 0.0483, "depth": 0.3}	m	published + estimate	1-1/2 in Schedule 40 PVC irrigation supply line, OD 48.3 mm (ASTM D1785); 0.30 m cover (irrigation practice, estimate)
PAIL	{"r": 0.12, "height": 0.25, "wall": 0.003, "fill": 0.15}	m	retailer + estimate	2-gallon HDPE pail ~9.5 in across, 10 in tall (estimate from listings); bedding sand 15 cm deep (~8 kg)
STONE_RHO	2650.0	kg/m3	published	quartz/granite fieldstone
TARP	{"size": [1.83, 2.44], "mass": 0.4}	m, kg	retailer	6 x 8 ft poly tarp
POST_TEST	{"force": 150.0, "height": 1.0, "ramp_s": 0.5, "hold_s": 0.5, "creep_s": 2.0, "creep_window_s": 0.25, "unload_	N, m, s, s, s, s, s, s, s, m, deg, deg	published + authored	set-post verdict (task.Evaluator.post_load_test), on a restored copy of the state with the unburied tools parked and the braces released. Load (authored estimate): a person leaning on a post, 150 N at 1 m above grade, ramped over 0.5 s and held 0.5 s, toward +x, -x, +y and -y. Reading: the peak horizontal motion of the post at grade during the push, and that motion projected 2
POST_FINISH	{"low": -0.03, "crown": 0.05, "share": 0.9, "yard_low": -0.03, "yard_share": 0.95, "lip_inset": 0.015, "lip_ma	m, m, -, m, -, m, m, m, m, m, m, N, -, m3	authored	set-post finish, read from 1 cm column tops (task.Ledger.heightfield). Backfill to grade: >= 90 % of the dug footprint's columns (inside the hand-dug lip outline at grade less 15 mm, the post section + 10 mm excluded) within -30/+50 mm of grade: 'back up to grade' with a slight drainage crown; a mean would let pits cancel heaps. The outermost 15 mm inside the lip and the first
BRACE	{"k": 20000.0, "c": 150.0, "height": 1.0, "reach": 0.9}	N/m, N s/m, m, m	estimate	two 1x2 diagonal braces nailed to the post 1 m up and to stakes 0.9 m out: axial stiffness dominated by the nailed joints (estimate)
STRIKE	{"force_n": 30.0, "speed_m_s": 0.2}	N, m/s	authored	a tool contact on the marked lateral with normal force > 30 N or approach speed > 0.2 m/s counts as a strike to report (a light trowel touch locates the line; Nebraska 811 hand-dig rules [811])

8. References

The source keys cited in the parameter ledger.

[AM1] Ames Razor-Back 47171 trenching shovel (vamac, homebyames, ammc listings).

[AM2] Ames/True Temper post-hole digger catalogue (M.E. Campbell PDF) and listings.

[BI] R. A. Birch, E. I. Ekwue, C. J. Phillip, "Soil-metal sliding resistance forces of some Trinidadian soils at high water contents", West Indian J. Eng. 38(2) (2016), Table 3.

[CAT swell tables] Swell/shrink tables (Caterpillar Performance Handbook values via Quantum Land Design, NORDIX, pdfcoffee swell table); Caterpillar, swell and void percentage load factors, https://www.cat.com/en_US/articles/ci-articles/swell-and-void-percentage-load-factors.html

[DW] DeWalt DCD996 data sheet, distributor-hosted manufacturer document, https://assets.cef.co.uk/downloads/pdg/dewalt_dcd996p2-gb_datasheet/dewalt_dcd996p2-gb_datasheet.pdf

[FI] Fiskars Ergo trowel listings.

[IT] INTERTOOL/ROK 1.5 lb mini mattock listings.

[PP] Power Planter 312 (Lowe's, Home Depot, Hartville, Hardware World listings); Power Planter catalogue, https://powerplanter.com/collections/planting-annuals

[PSU] Penn State Extension, "Diagnosing soil compaction using a penetrometer".

[SJ] Spear & Jackson 4450DS/1160SP spade pages; Traditional Stainless digging spade product page, https://www.spear-and-jackson.com/products/garden-tools/traditional-stainless/digging/traditional-stainless-digging-spade

[Z60] AmericanHort, ANSI Z60.1 (2025/2026), Table 1 container classes.

[811] Nebraska811 excavator manual (with the Iowa One Call 2024, Ohio811 2026 and Montana811 manuals).